

Index Terms

Duty cycle constraint, capacity-achieving input, mutual information, entropy rate, Markov process, hidden Markov process (HMP), Monte Carlo method.

I. INTRODUCTION

In many wireless communication systems, a radio is designed to transmit actively only for a fraction of the time, which is known as its *duty cycle*. For example, the ultra-wideband system in [1] transmits short bursts of signals to trade bandwidth for power savings. The physical half-duplex constraint also requires a radio to stop transmission over a frequency band from time to time if it wishes to receive useful signals over the same band. Thus wireless relays are subject to duty cycle constraint, so do cognitive radios which have to listen to the channel frequently to avoid causing interference to primary users. The *de facto* standard solution under duty cycle constraint is to transmit packets intermittently.

This work studies the fundamental question of what is the optimal signaling for a Gaussian channel with duty cycle constraint as well as average transmission power constraint. An important observation is that the signaling in nontransmission periods can be regarded as transmission of a special *zero* signal. We first make a simplistic and idealized assumption that the analog waveform corresponding to each transmitted symbol is exactly of the span of one symbol interval. We restrict our attention to discrete-time scalar additive white Gaussian noise (AWGN) channels for simplicity, where the duty cycle constraint is equivalent to a requirement on the minimum fraction of zero symbols in each transmitted codeword, which is called the *idealized duty cycle constraint*. We then consider the case where a practical pulse shaping filter is used, e.g., for band-limited transmissions. As such, during a transition between a zero symbol and a nonzero symbol, the pulse waveform of the nonzero symbol leaks into the interval of the zero symbol. A realistic duty cycle constraint must include the extra cost incurred upon transitions between zero and nonzero symbols. The mathematical model of the preceding input-constrained channels is described in Section II.

Determining the capacity of a channel subject to various input constraints is a classical problem. It is well-known that Gaussian signaling achieves the capacity of a Gaussian channel with average input power constraint only. In addition, Zamir [2] shows that the mutual information rate achievable using a white Gaussian input never incurs a loss of more than half a bit per sample with respect to the power